

Final Exam of Computational Neuroscience Course 2025

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There are *four* problem sets in the Final Exams. You are free to pick any *three* of them. The last problem set is an open project without any actual answers.

Limit cycle behavior with two neurons

Let us consider a nonlinear network with a limit cycle attractor with two neurons:

$$\begin{aligned}\tau \frac{dx}{dt} &= -x + \tanh(Jx - Ky) \\ \tau \frac{dy}{dt} &= -y + Gx\end{aligned}\tag{1}$$

where J , K , and $G > 0$ are constants and τ is the time constant of the neurons. Note that x is an excitatory neuron and y is an inhibitory neuron. It is thus, a very generic model consisting of coupled activator and suppressor units. The system shows a variety of behaviors, including a stable limit cycle.

- a. Find the fixed points of the system. Find the set of parameters for which each fixed point exists.
- b. Consider the fixed point $(x, y) = (0, 0)$. Setting $\tau = 1$, sketch the phase diagram of the fixed point, plotting J against KG . In the $J - KG$ plane, draw and label the following four regions:
 - i. The fixed point is stable and the system converges monotonically to the stable state under small perturbation.
 - ii. The fixed point is stable and the system spirals back to the stable state under small perturbation.
 - iii. The fixed point is unstable and the system diverges monotonically away under small perturbation.

- iv. The fixed point is unstable and the system spirals away under small perturbation.
- c. Consider the case in which $(x, y) = (0, 0)$ is unstable and the other fixed points do not exist. Find a closed surface in the x - y plane along which all the flow points inwards. Qualitatively describe how the system evolves at long times in this case, given initial conditions inside the surface you found.
- d. Draw a new phase diagram in the $J - KG$ plane for the entire system, classifying when the system has 1 stable fixed point, when it has 2 stable fixed points, and when it produces a limit cycle.
- e. Numerically integrate this system of equations for J , K , and G parameters representative of each of the four cases in (b), one of which will also be the case discussed in (c). Show how the system evolves in each case starting from an initial condition near $(0, 0)$. Make sure you integrate long enough to see a steady state behavior emerge. (You should hand in graphs of the x - y plane. For your own edification, plot $x(t)$ vs. t and $y(t)$ vs. t to make sure the system has reached a steady state. If you use Matlab, `ode45` is a useful integration command).

Fisher Information and Mutual Information

Consider a group of N cells responding to a stimulus x . The response of each neuron i is given by $r_i = wx + z_i$ where z_i varies from trial to trial as a gaussian random variable with zero mean and variance σ_i^2 . The z_i are uncorrelated: $\langle z_i z_j \rangle = \delta_{i,j}$. The stimulus x is drawn from a gaussian distribution with zero mean and variance σ_0 .

- (a) Compute the Fisher Information of the system.
- (b) Recall what I discussed in the class about the Maximum Likelihood (ML) estimate

$$\operatorname{argmax}_x p(r|x).$$

Give an expression for the ML estimator of $\hat{x}$. Evaluate the mean square error of the estimate $\langle (x - \hat{x})^2 \rangle$, and compare with the result of (a).

- (c*) Instead of using ML estimate, now let us consider the Bayes rule:

$$p(x|r) = \frac{p(r|x)p(x)}{p(r)}. \quad (2)$$

We would like to find a Bayesian estimator $\hat{x}$ that would maximize the posterior distribution $p(x|r)$. Evaluate its mean square error and explain its dependence on σ_0 . Compare your results with (b) and Under what conditions you expect

that the two estimators will differ significantly? Explain the relationship between the results and the Cramer-Rau Bound.

(d) Compute the Mutual Information between the neuronal responses and the stimulus. What is the relationship between your result and the Fisher Information? Assume now that all $\sigma_i = \sigma$. How do both quantities depend on the population size.

Guide: In the above, the mean square error average $\langle \dots \rangle$ is defined as an average over both neuronal noise and stimulus values.

Associative Memory with Decaying Memory Traces

Consider a network of binary neurons with the dynamics similar to that of the Hopfield Model.

Network dynamics: The value of each neuron is represented by $S_i = \pm 1$. The input function to each neuron is defined by $h_i(t)$, which stands for the weighted sum of the values of the rest of the neurons in the networks, weighted by the connection matrix W :

$$h_i(t) = \sum_{j=1}^N W_{ij} S_j(t) \quad (3)$$

At each time step a single neuron is chosen and its state is updated according to the following rule:

$$S_i(t + \Delta t) = \text{sgn}(h_i(t)) \quad (4)$$

Connectivity Matrix: The memories are stored in a temporal sequence by updating the weight matrix W_{ij} dynamically. We start with $W_{ij} = 0$. As time proceeds, we add a new stored pattern ξ^μ to the existing memory while allowing the other memories to decay exponentially:

$$W_{ij}^t = \lambda W_{ij}^t + \frac{1}{N} \xi_i^\mu \xi_j^\mu \quad (5)$$

($W_{ii} = 0$). We let $\lambda = e^{-\tau}$, $\tau > 0$. (This kind of memory is dubbed a *palimpsest* — papyrus that has been written on more than once, with the earlier writing incompletely erased and often legible.)

- Show that the connection matrix can be written as:

$$W_{ij}^p = \frac{1}{N} \sum_{\mu=0}^p e^{-\mu\tau} \xi_i^\mu \xi_j^\mu, i \neq j \quad (6)$$

- Consider a steady state situation where the time step p approaches infinity. Estimate how many past memories we can store, assuming all ξ_i^μ are drawn independently with equal probability of ± 1 .

Hint: To estimate the memory capacity, use a ‘signal-to-noise’ analysis similar to that presented in the lecture for the Hopfield Model. A pattern ξ^μ is stable if for all neurons $\xi_i^\mu h_i^\mu > 0$.

1. Derive an expression for the mean and variance of this quantity.
2. Assuming that N and p large, approximate the distribution of $\xi_i^p h_i^p$ by a normal distribution. Find the maximum p for which the probability of error (instability of the stored pattern) is small. Note: For the sake of calculation, it is useful to define new variables $\bar{\tau}$ and $\bar{p}$ with $\tau = \frac{\bar{\tau} \ln N}{N}$, $p = \frac{\bar{p} N}{\ln N}$, (so that in particular p is expected to be large and τ is expected to be small) to capture the expected dependence on N .
3. Plot the maximal value of $\bar{p}$ as a function of $\bar{\tau}$ and discuss the results and their implications on the performance of this model in comparison to the Hopfield model.

Optimal Receptive Field in 1-Dimensional World

We will follow our lecture on the efficient coding principle, which allows us to derive the optimal synaptic weight W (also called receptive field for the output neurons) that maximizes the mutual information in the presence of noise. Make sure you understand the lecture note I distributed. Now imagine a 1-dimensional visual world on a circle with inputs that are correlated according to

$$c(n,m) = \langle x(n)x(m) \rangle = c(n-m) = \exp\left(-\frac{|n-m|}{D}\right), \quad (7)$$

where n is a spatial index running from 1 to N (note: with periodic boundary conditions, $|n-m|$ for distances larger than $N/2$ should be replaced by $N - |n-m|$). The input contains additional spatially uncorrelated input noise with variance σ_z^2 . We want to find the shape of the receptive fields, namely the functional form of W that are optimized for this world.

- (a) Using my lecture note as a guide, show that in the spatial frequency domain, the optimal receptive field (or filter) $w(k)$ is given by the equation

$$w^2(k) = \frac{\sigma_\eta^2}{\sigma_z^2} \left(\frac{1}{2} \left[\frac{c(k)}{c(k) + \sigma_z^2} \left(1 + \sqrt{1 + \frac{\mu}{c(k)}} \right) \right] - 1 \right),$$

where $c(k)$ is power spectrum of the visual inputs, σ_η^2 is the output noise. We will set $\sigma_\eta^2 = 1$ in this model. Please recall my lectures what is a power spectrum and how it is related to the eigenvalues of the input covariance matrix, which happens to be a circulant matrix in our 1-dimensional visual world. μ is a term involving the lagrange multiplier λ , more precisely $\mu \sim 1/\lambda$.

- (b) As mentioned in the lecture, let us make a specific requirement that the total variance (or power) of the outputs $\mathbf{y}$ is independent of input noise, and is set as a constant. Find the dependence of $w(k)$ on $c(k)$ analytically, in the limits of small and large input noise σ_z^2 . Interpret your results. **Guide:** In order to derive these limits, you will need to consider how μ varies with σ_z^2 so that the total output variance remains a constant. It can be shown that for small noise $\mu \propto \sigma_z^2$, where as for large noise, $\mu \propto \sigma_z^4$.
- (c) Numerically Fourier transform $W(k)$ to calculate the spatial shape of the optimal filters. Use $N = 401$ and $D = 30$ and study different values of noise, σ_z^2 . Qualitatively explain your results, by plotting the spectrum $W(k)$ vs. k and $W(n)$ vs. n for different values of input noise. Compare these numerical results with your analytical predictions from (b). You will find the matlab functions `fft` and `ifft` useful.